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REVIEW ARTICLE

Enrichment of antibiotic resistance genes (ARGs) in polyaromatic hydrocarbon–contaminated soils: a major challenge for environmental health

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Abstract

Polyaromatic hydrocarbons (PAHs) are widely spread ecological contaminants. Antibiotic resistance genes (ARGs) are present with mobile genetic elements (MGE) in the bacteria. There are molecular evidences that PAHs may induce the development of ARGs in contaminated soils. Also, the abundance of ARGs related to tetracycline, sulfonamides, aminoglycosides, ampicillin, and fluoroquinolones is high in PAH-contaminated environments. Genes encoding the efflux pump are located in the MGE and, along with class 1 integrons, have a significant role as a connecting link between PAH contamination and enrichment of ARGs. The horizontal gene transfer mechanisms further make this interaction more dynamic. Therefore, necessary steps to control ARGs into the environment and risk management plan of PAHs should be enforced. In this review, influence of PAH on evolution of ARGs in the contaminated soil, and its spread in the environment, has been described. The co-occurrence of antibiotic resistance and PAH degradation abilities in bacterial isolates has raised the concerns. Also, presence of ARGs in the microbiome of PAH-contaminated soil has been discussed as environmental hotspots for ARG spread. In addition to this, the possible links of molecular interactions between ARGs and PAHs, and their effect on environmental health has been explored.

Keywords Antibiotic resistance · Class 1 integrons · Horizontal gene transfer · Stress response · PAH contaminations · Resistome · Environment sustainability

Introduction

Antibiotic resistance genes (ARGs) and antibiotic-resistant bacteria (ARB) are being considered emerging contaminants and serious environmental problems. Therefore, the scientific community is now gradually exploring the role of the environment as a source and dissemination route for antimicrobial resistance (AMR) (Berendonk et al. 2015). The cumulative frequency of AMR in this era recommends an emergent worldwide human health concern; as it has been stated that by 2050, worldwide yearly mortality is anticipated to be 10 million if an emergent action plan is not designed and implemented instantly to combat AMR (de Kraker et al. 2016).

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Polyaromatic hydrocarbons (PAHs) are the group of semi-volatile organic compounds which are highly toxic, with significant concerns because of their mutagenic and carcinogenic effects, and genotoxic properties (Montaño-Soto and Garza-Ocañas 2014). PAHs refer to compounds with two or more benzene rings joined in undeviating, or bunch, or pointed arrangements (Gorovtsov et al. 2018; Cunningham et al. 2020a). They are usually distributed in an extensive range of tissues with a noticeable affinity for localization in body fat and are freely absorbed in the gastrointestinal tract of mammals due to their high solubility in lipid (Alegbeleye et al. 2017; Jiang et al. 2020). PAH contaminations may lead to mutation, cancer, DNA damage, and lesion in the liver, in addition to several other abnormalities (Reid et al. 2016).

The development of AMR in PAH-contaminated soils is now being considered a serious challenge (Chen et al. 2017). PAHs, in the environment, are continuously increasing extensively through anthropogenic sources due to cumulative effects of industrial activities and suburbanization (Shokralla et al. 2012). The struggle to manage this problem goes beyond the restriction of the presence of PAHs in a normal environment and/or uncontrolled use of antibiotics in healthcare and agriculture sectors leading to their discharge into the environment. PAHs are widely present contaminants of terrestrial and aquatic environments (Ben Said et al. 2008; Chen et al. 2017; Gorovtsov et al. 2018), which are discharged along with other hydrocarbons either through human activities or unintentionally, causing the disturbances of the biological cycle in between natural environment and living being.

ARGs, ARBs, antibiotics, and other toxic substances are found to be present in the soil in a wide range of the environment such as wastewater treatment plants, ocean sediments, sewage, hospital effluents ground, and surface water (Table 1) (Ben Said et al. 2008; Li et al. 2015; Wang et al. 2017; Chen et al. 2017; Jang et al. 2018; Maruzani et al. 2018a; Muccee and Ejaz 2019). There are many reports of ARGs present in natural environments, which act as a site for their enrichment (Table 1) (Versluis et al. 2015). Even wetlands have been reported to have ARGs, and isolates such as *Arcobacter* sp., from the urban wetlands, have been reported to be resistant to multiple antibiotics such as tetracycline and sulfonamides (Hsu et al. 2017). Polluted soil not only influences the ARGs but also influences the production of antibiotics by the indigenous soil bacteria, and this may increase the selective pressure, which leads to the proliferation of ARBs (Hemala et al. 2014). Furthermore, ARGs and other resistance genes (e.g., PAHs/heavy metal) are often present in the mobile genetic elements (MGEs) (Knapp et al. 2011).

There are previous reviews on antibiotic resistance and its origin, ecology, and dissemination potential in soil resistome (Nesme and Simonet 2015), in estuarine sediments (Rodgers et al. 2019). Also, there had been discussions on ecological roles of bacterial MDR efflux pumps in microbial ecosystems, including non-clinical environments contaminated with toxic compounds (Martinez et al. 2009). Potential risks of ARBs and ARGs in hydrocarbon-contaminated soil has also been reviewed (Cunningham et al. 2020b). The possible underlying physiological and molecular mechanisms for the development of ARGs in PAH-contaminated environments also need attention. Therefore, in this article, we have focused on the mechanisms through which PAH contaminations may enrich the ARGs in the polluted soil, and also elaborated the molecular mechanisms involved during interactions of PAHs with ARGs, with gene transfer mechanism's, and relevant physiological attributes in the contaminated areas.

The occurrence of ARGs in PAH degraders

There are several reports where antibiotic resistance has been reported from PAH degraders. Stancu and Grifoll (2011) observed six gram-positive isolates belonging to genera *Bacillus*, *Lysinibacillus*, and *Rhodococcus* and eight gram-negative strains belonging to the genera *Aeromonas*, *Klebsiella*, *Pseudomonas*, and *Shewanella* to be tolerant to naphthalene, 2-methyl naphthalene and fluorine (Table 1) (Stancu and Grifoll 2011). In the presence of hydrophobic antibiotics like ampicillin and kanamycin, the resistance of these bacteria to toxic compounds was affected. Also, gram-positive bacteria were more resistant to a toxic substance as their cellular and molecular modifications were being influenced by antibiotics (Stancu and Grifoll 2011).

Pseudomonas veronii CSGG7 and *Yersinia inetrmedia* CSGN3 belonging to γ-Proteobacteria had been observed to degrade hydrocarbon and resist antibiotic and heavy metal. Furthermore, it was observed that *P. syringae* strain CSGN4 and *P. mandelii* CSGN1, which were tolerant to cefoperazone, cefuroxime sodium, ceftazidime, cefaclor, cefotaxime, moxalactam, tigecycline, netilmicin, piperacillin, amoxicillin, novobiocin, piperacillin-tazobactam, norfloxacin, imipenem, and nalidixic acid had lesser efficiency for the degradation of hydrocarbons (Mathe et al. 2012). *P. aeruginosa*, *Acinetobacter baumannii*, and *Klebsiella pneumoniae* isolated from contaminated soil from garages and petrol pumps of Delhi, India, were found to be resistant to ampicillin, tetracycline, kanamycin, and rifampicin. These isolates were able to remediate PAHs from the contaminated environments (Bhagat et al. 2019). However, Mathe et al. (2012) studied the ability of hydrocarbonoclastic bacteria for bioremediation of aromatic hydrocarbon–contaminated sites in the presence of antibiotics. They revealed that members of the genus *Pseudomonas* belonging γ-Proteobacteria were dominant in PAH-contaminated sites (Mathe et al. 2012). In the case of Balan-contaminated site (polluted with heavy metal and hydrocarbons), bacterial strains exhibited strong relation among hydrocarbon degradation ability and antibiotic resistance, since resistance and catabolic genes were located on a plasmid, and shared within the bacterial community by horizontal gene transfer (HGT). However, there was no correlation among degradation ability and antibiotic resistance possibly due to the slow rate of transmission of mobile genetic elements in the Sandominic-contaminated site which was solely polluted with PAHs (Mathe et al. 2012).

The plant-associated bacteria used in rhizoremediation of PAHs had also been reported to have ARGs. An endophytic bacterium *Enterobacter* sp. 12J1, isolated from plants, grown in PAH-contaminated soils (near an oil refinery) was able to degrade pyrene (83.8%) at 28 °C experimented for 7 days in pots amended with pyrene (5 mg/l). It could colonize tissue and rhizosphere of wheat and maize and produced indole acetic acid (IAA), siderophore, and solubilize inorganic phosphate. The isolate showed resistance to streptomycin (10 μg/ml), kanamycin (10 μg/ml), spectinomycin (10 μg/ml), rifampicin (20 μg/ml), and ampicillin (100 μg/ml) (Sheng et al. 2008). Another endophytic bacterium *Massilia* sp. Pn2 isolated from *Alopecurus aequalis* Sobol grown in PAH-contaminated soils could degrade 95% of phenanthrene (150

Table 1 Presence of ARGs, ARBs, and MGEs/integrons in PAH-contaminated areas

Sl. no.	Samples/PAHs	ARBs	ARG types	Antibiotic categories	Integrons/MGEs	References
1	Petrol-contaminated soil	*Brevibacillus agri*, *Burkholderia pyrrocinia*, *Burkholderia lata*, *Brevibacillus formosus*	*mecA*, *CmR*	Chloramphenicol, macrolide	--	(Muccee and Ejaz 2019)
2	Pyrene	Sphingobium sp.	*tetM, tetW, sulI, sulIII*	Tetracycline	Class 1	(Sun et al. 2015)
3	Benzo[a]pyrene, pyrene, phenanthrene	*Shigella flexneri*	*tet*^{R}, *tet*^{s} *aadA1*	Tetracycline	--	(Maruzani et al. 2018b)
4	Pyrene, benzo[g,h,i] perylene phenanthrene	*E. coli*	*AmpR*	Ampicillin	--	(Kang et al. 2015)
5	Naphthalene, phenanthrene	*Proteus mirabilis*, *Pseudomonas monteilii*, *Aerococcus*	*intI1*, *sulI*, *aadA2*	--	Class 1 integrons	(Wang et al. 2017)
6	Environmental samples from multiple ecosystems	*Escherichia coli*, *Klebsiella pneumonia*, *Pseudomonas aeruginosa*, *Salmonella enterica*, *Acinetobacter baumannii*	*intI1*, *sul1*, *tetA*, *tetB*, *tetC*, *VEB-6*, *VEB-1*, *e*, *dfrA17-aadA5*, *aadA1*, *aadA2*, *aadA4*, *aadA5*, *aacA29b*, *qacE*	Tetracycline	Class 1 integrons	(Ma et al. 2017)
7	PAHs released from anthropogenic sources (e.g., oil spill, industrial runoff)	Gammaproteobacteria, Proteobacteria, Bacteroidetes	*acrA*, *acrB*, *amrB*, *bacA*, *ceoB*, *macB*, *mexB* and *tolC*, *amrB*, *bacA* and *ceoB*		--	(Chen et al. 2017)
8	Effluent of coastal aquaculture	Vibrio, Marinomonas, Gammaproteobacteria, Proteobacteria, Bacteroidetes, Flavobacteriia, Pseudoalteromonas	*tetA*, *tetB*, *tetD*, *tetE*, *tetG*, *tetH*, *tetM*, *tetQ*, *tetX*, *tetZ*, *tetBP*, *sul1*, *sul2*	Tetracycline	Class 1 integrons	(Jang et al. 2018)
9	Wastewater treatment plants	--	*aacA4-ereA1*, *orf-aadA2-bla*$_{OXA\text{-}129}$*-linF*, and *catB8-bla*$_{OXA\text{-}1}$*-aadA1*, *aacA4*, *cat_1*, *emrE*, *aacA4*, *ant1*, *dfrA*, *dhfrI*		Class 1, class 2, and class 3 integrons	(An et al. 2018)
10	Pyrene	*Acinetobacter* sp., *Kocuria* sp.	*KanR*, *AmpR*, *ErmA*, *AacA4*	Kanamycin, ampicillin, erythromycin	--	(Sun et al. 2014)
11	Anthropogenic pollutants	--	*tetO*, *tetM*, *tetW*, *tetT*, *tetQ*, *tetB/P*	Tetracycline	Class 1 integrons	(Zhou et al. 2017)
12	Oily sludge	*Bacillus*, *Shewanella*, *Lysinibacillus*, *Aeromonas*, *Rhodococcus*	*AmpR*, *KanR*	Kanamycin, ampicillin	--	(Stancu and Grifoll 2011)
13	Heavy oil–contaminated soil	Gammaproteobacteria, Actinobacteria, Alphaproteobacteria	*TetR*, *sul1*	Tetracycline	--	(Hemala et al. 2014)
14	Fluoranthene/sediment-contaminated with PAHs	Gammaproteobacteria	*TetR*, *AmpR*, *Sul1*, *qacE*	Tetracycline, ampicillin	--	(Ben Said et al. 2008)
15	Livestock wastes applied to agricultural soil	*Salmonella*, *Listeria*, and *Campylobacter* spp. and *Escherichia coli* O157	*qnr*, *sul1*, *qacE*		Class 1 integrons	(Hutchison et al. 2004)
16.	Petroleum-contaminated soil samples	*Pseudomonas* sp., *Burkholderia* sp., *Xanthomonas* sp., *Klebsiella* sp., *Salmonella* sp., *Stenotrophomonas* sp., *Mycobacterium* sp., *Desulfitobacterium* sp., *Alkaliphilus* sp., *Clostridium* sp., *Bacillus* sp., *Legionella* sp., *Alteromonas* sp., *Enterococcus* sp., *Rhodopseudomonas* sp., *Bradyrhizobium* sp., *Methylobacterium* sp., *Methylibium* sp., *Methylococcus* sp., *Nitrococcus* sp., *Paracoccus* sp., *Acidovorax* sp., *Aromatoleum* sp., *Bacteroides* sp., *Erythrobacter* sp., *Leptothrix* sp., *Methylibium* sp., *Chromobacterium* sp.	*CTX-M-16*, *VanR*, *MexT*, *PmxA*, *FosX*, *AmpS*, *MacA*, *MexD*, *CmeA*, *CmeC*, *MdtC*, *VanW*	Cefotaxime, vancomycin, fosfomycin, macrolide, polymyxin	Class 1 integron, class 2 integron	(Das et al. 2020)

mg/l) within 48 h. The isolate was tested for antibiotic resistance to some antibiotics (ampicillin, chloramphenicol, kanamycin, rifampicin, streptomycin, and tetracycline) and found to be resistant to ampicillin (75 mg/l) and chloramphenicol (25 mg/l) (Liu et al. 2014). Endophytic *Acinetobacter* sp. BJ03 and *Kocuria* sp. BJ05 isolated from plants grown in PAH-contaminated soil near a petrochemical plant were found to improve the degradation of pyrene. These bacteria were resistant to ampicillin, chloromycetin, erythromycin, gentamycin, kanamycin, and spectinomycin (Sun et al. 2014). Kang et al. (2017) suggested that bacteria belonging to the genus *Pseudomonas* identified as PGPR strains in their study were found to possess multiple ARGs; therefore, attention should be given on ARGs for use in agricultural soil for improvement of crop production (Kang et al. 2017). *P. aeruginosa*, *Acinetobacter baumannii*, and *K. pneumoniae* isolated from contaminated soils of garages and petrol pumps of Delhi, and NCR region, were reported as potent degraders of PAHs, and they are resistant to ampicillin, kanamycin, rifampicin, and tetracycline (Bhagat et al. 2019).

The occurrence of ARGs in the microbiome of PAH-contaminated soil

With the gradual increase in number of reports of ARGs from PAH-contaminated soils, the latter is now being realized as one of the major reservoir of ARBs and ARGs (Nesme and Simonet 2015). Large varieties of ARGs have been frequently discovered in the soils for major resistance mechanisms, which largely expanded our understanding of antibiotic resistome in PAH-contaminated soils (Chen et al. 2017). However, investigation of archived soils confirmed that ARG abundance has substantially increased since the antibiotic era began in 1940 (Graham et al. 2016). High level of ARGs in polluted areas marked the high stress of anthropogenic pollutants that support the normal opinion that AMR only takes place because of the high use of antibiotics (Table 1) (Chen et al. 2017). Due to the discharges of toxic substances and organic hydrocarbons from the industries into the sewage, the sewage sludge became the chief reservoir for the fluoroquinolone (Golet et al. 2003). Reports from some field trials for use of slurry into the agriculture field confirmed the persistence of fluoroquinolones in the treated soils which also suggested the inadequate movement of fluoroquinolones into the subsoil (Nordmann and Poirel 2005). Hutchison et al. (2004) reported that since quinolone resistance genes (*qnr*) are present on class 1 integrons that carry ESBL gene types, therefore, determination of fluoroquinolone resistance became significant as they co-select class 1 integrons and ESBL genes (Table 1) (Hutchison et al. 2004).

Metagenomic profiling of emergence and diversity of ARGs under PAH stress demonstrated that ARG abundance was 15 times higher in contaminated sites. Antibiotic-resistant bacteria belonging to proteobacteria were observed in PAH-contaminated soil near the wastewater pond of the petrochemical plant. More than 10 groups of ARGs for beta-lactams (penicillin, ampicillin, cefazolin, cefoxitin, and ceftriaxone) chloramphenicol, aminoglycosides, macrolides, fluoroquinolones, tetracyclines, and polypeptides had been identified in PAHs-contaminated soils, which also include ARG subtypes for aromatic antibiotic transporters through efflux pumps (Chen et al. 2017). Also, PAHs were observed to play a key role as a substrate for efflux transporters consisting of *acrA*, *acrB*, *tolC*, and *amrB* (Chen et al. 2017). In another report, a total of 130 PAH-degrading bacterial isolate, out of which 75% belonged to Gammaproteobacteria, could utilize fluoranthene as a sole source of carbon, and almost all the strains were observed to be resistant to beta-lactam and aminoglycosides. Fifteen percent of the bacterial strains were found to be resistant to neomycin and 97% to oxacillin. This was indicative of the selection of PAHs on antibiotic resistance genes carrying bacteria accumulated in the soil (Ben Said et al. 2008).

Soil microbiomes are influenced by PAHs leading to an increase in the population of Actinobacteria, which are known for antibiotic formation possessing multiple ARGs. It has been presumed that soils contaminated with PAHs could be a selective environment for ARBs and ARGs could be transferred to the pathogenic bacteria. Furthermore, it has been suggested that properties of the soil inclined the bioavailability of PAHs and its concentration that can give an improper estimation of the contamination which developed antibiotic resistance (Gorovtsov et al. 2018)

The relationship between antibiotic resistance profile and taxonomic affiliation of the bacterial strains in PAH-contaminated soil suggests that genera from Actinobacteria and Gammaproteobacteria harbored resistance against multiple antibiotics. Forty-nine percent of the bacterial strains were observed to be resistant to penicillin, 28% to chloramphenicol or rifampicin, 26% to streptomycin, and 9% to tetracycline out of 47 bacterial strains that were isolated from an industrial alpine site contaminated with hydrocarbons (Hemala et al. 2014).

The expression of ARGs in PAH-contaminated soil was found to be high in the rhizosphere of willow plants during the phytoremediation process (Yergeau et al. 2014). The abundance of sulfonamide resistance gene (*sulI*), aminoglycosides resistance gene (*aadA2*), and class I integrase gene (*intI1*) were found to be influenced by the presence of different concentrations of naphthalene and phenanthrene in coastal microbial communities. One hundred milligrams per liter of naphthalene and 10 mg/l of phenanthrene concentration resulted in the enhancement of abundance of these ARGs (Wang et al. 2017). Further, it was observed that class I integron–mediated conjugative transfer was mainly responsible for a higher abundance of ARGs.

Antimicrobial resistance and PAH tolerance as coexisting traits in bacteria

It was suggested that the interaction of mutagenic compounds at the cellular level is important for the development of ARGs in the bacteria in the normal environment (Wellington et al. 2013). Due to HGT in the soil microorganisms, the plasmids harboring diverse ARGs may come in contact with PAHs (Jussila et al. 2007; Kang et al. 2010), and such interactions between plasmids and PAHs are governed by non-covalent interactions (Kang et al. 2015) in the absence of the natural enzymes. Fluorescence studies confirmed that phenanthrene and pyrene can bind with the genetic bases and block the biological enzymolysis of DNA (Kang et al. 2010). These effects were common for phenanthrene and less common for the pyrene. However, in the case of benzo-a-pyrene, there were no distinct effects on transformation frequencies (Kang et al. 2015).

Also, it is significant to mention that, in the contaminated environmental systems, organic co-contaminants/anthropogenic substances other than PAHs may enrich the ARGs by the anti-hydrolysis mechanisms, and non-covalent binding of genetic bases (Kang et al. 2015; Shou et al. 2019). Therefore, a positive correlation between PAHs and ARGs may not necessarily mean the causal interdependence between these two, as other factors, such as presence of co-contaminants may also have some role. For instance, in some sites with more PAH contaminants, there may be more other organic pollutants (e.g., substituted aromatic hydrocarbons). There is still a gap in understanding of molecular mechanisms due to lack of laboratory evidence. Shou et al. (2019) reported that coexisting substituted aromatic hydrocarbons could facilitate ARGs to cross the extracellular polymeric substances (EPS) and reach inside the cells, and thereby enabling the dissemination of ARGs. Although, the mechanism is still unknown, the authors demonstrated that anti-hydrolysis mechanism of mobile plasmid in the extracellular matrix contributes and increases the chances of dissemination of ARGs (Shou et al. 2019).

Campbell et al. (1995) had reported that *Pseudomonas* has resistance to heavy metals and antibiotics along with PAH tolerance. The isolates from industrial soils could tolerate PAHs and heavy metals, and were also resistant to multiple antibiotics like ampicillin, chloramphenicol, cephalothin, erythromycin, novobiocin, and trimethoprim than in agricultural soils (Campbell et al. 1995). The study also suggested that antibiotic resistance in *Pseudomonas* could be plasmid-mediated as evidenced by *P. fluorescens* NR58 (streptomycin-resistant strain) that harbored a 58-kb plasmid (Campbell et al. 1995).

In microorganisms, resistance-nodulation-division (RND) efflux pumps are presumed to be responsible for the expression of multidrug resistance, pathogenesis, environmental adaptability, and other physiological processes (Martinez et al. 2009). RND efflux pumps have been reported to be identified and annotated in bacteria that could utilize PAHs indicating its role in the degradation of PAHs (Hearn et al. 2003; Chain et al. 2006). An RND efflux pump, EmhABC identified in *P. fluorescens* LP6a, had been reported to remove antibiotics and PAHs, including phenanthrene, anthracene, and fluoranthene. Both PAHs and antibiotics like chloramphenicol act as a substrate for the EmhABC efflux system (Hearn et al. 2003; Adebusuyi and Foght 2011). Two RND efflux pumps (TtgABC and SrpABC) identified in *P. putida* B6-2 were studied to elucidate their functions in PAH degradation. TtgABC exhibited higher basal expression levels and contributed to antibiotic resistance while SrpABC was found to release toxicity produced by intermediates during degradation and showed antibiotic resistance under an artificial overproduction condition (Yao et al. 2017). Overproduction of TtgGHI of *P. putida* DOT-T1E was observed to efflux of certain antibiotics like ampicillin, chloramphenicol, erythromycin, norfloxacin, and tetracycline (Molina-Santiago et al. 2014).

The disruption of *emhB*, a gene of RND pump system responsible for efflux of PAHs, decreased the resistance of the bacteria to chloramphenicol and nalidixic acid; however, it did not affect the resistance to other antibiotics like erythromycin, trimethoprim, tetracycline, and streptomycin (Hearn et al. 2003). Sun et al. further (2015) suggested that pyrene-contaminated soils have a high abundance of integrin *int1*, and with the help of int1, genes coding for EmhABC efflux pumps could be inserted into the mobile genetic elements. Furthermore, PAH contamination had been reported to lessen the horizontal transfer of ARGs (Sun et al. 2015). Association of PAHs with antibiotic-resistant plasmids (pUC19) of *Escherichia coli*–containing ampicillin resistance gene (*ampR*) revealed non-covalent interaction between the ARGs and PAHs. Lower molecular weight PAHs like phenanthrene could bind effectively to the plasmid, forming a clew-like plasmid-PAH complex which resulted in the reduction of the transformation of the ampicillin-resistant gene. Transformation was less significant in higher molecular weight PAHs like pyrene and benzo-pyrene, which is indicative of a reduction of HGT in PAH-contaminated sample (Kang et al. 2015).

On the contrary, it was observed that PAHs affected the sensitization of the resistant strain of *Shigella flexneri* (tet^R) by reducing the growth of tetracycline-resistant strain in the presence of tetracycline. Based on their observation, it was speculated that (a) the resistant variants face the competition for similar substrates during detoxification (Maruzani et al. 2018b) and (b) expression of ARG cassettes is increased due to PAHs (Sun et al. 2015). Sun et al. (2015) also found that fluctuations in abundance of tetracycline resistance genes (*tetM*, *tetW*) and sulfonamide resistance gene (*sulI*, *sulIII*) were positively correlated with concentrations of pyrene in soil (Table 1) (Sun et al. 2015). Therefore, all these reports confirm the interlink between ARG and PAH contamination.

Absolute abundance of ARGs in polluted sites

It is significant to mention that in the contaminated environments, assessment of the absolute abundance of ARG provides more significant information, than relative abundance. Absolute abundance of ARGs is defined as the product of absolute abundance of 16SrRNA, and ratio of the relative copy number of ARGs and relative copy number of 16SrRNA (Li and Zhang 2020). There are few reports where absolute abundances of ARGs have been emphasized. Recently, Cheng et al. (2020) have reported that the absolute abundance of ARGs in the water, soil, and sediment samples was in the range of 5.96×10^7 to 6.69×10^9 copies l^{-1}, 4.27×10^8 to 2.64×10^{10} copies g^{-1}, and 2.62×10^8 to 8.67×10^{10} copies g^{-1}, respectively (Cheng et al. 2020). While in subtropical urban reservoir, the absolute abundance of ARGs ranged from 4.03×10^6 to 3.72×10^9 copies/l, whereas in case of MGEs, it increased from 6.62×10^5 to 1.84×10^9 copies/l (Fang et al. 2019).

High values of absolute abundance of ARGs had been reported from natural water bodies (i.e. in lake, river, and sea). The absolute abundance of *ermB* genes (macrolide resistance) in seawater was 8.61×10^7 copies/l, while that of *rpoB* and *katG* genes (anti-tuberculosis resistant) in the Haihe River Basin was 1.32×10^6 copies/l and 1.06×10^7 copies/l, respectively (Su et al. 2020). However, in anthropogenically impacted river such as the Weihe River, absolute abundance of ARGs was found to differ in different samples as follows: biofilm (2.11×10^7 to 5.27×10^9 copies/g) > sediment (4.31×10^7 to 4.73×10^8 copies/g) > water (3.04×10^4 to 8.41×10^5 copies/l) (Li and Zhang 2020). In other reports, it had been found that, in estuaries, absolute abundance of ARGs (vancomycin resistance genes) was 1.48×10^6 copies per gram (Zhu et al. 2017); while in surface water samples, absolute abundance of ARGs (vancomycin resistance gene) ranged from 4.44×10^6 to 8.17×10^8 copies per l (Zhou et al. 2017). Whereas in sand settling reservoirs and drinking water treatment plants across the Yellow River, China, total absolute abundance of ARGs in river water ranged from $1.51 \times 10^4 \pm 1.49 \times 10^4$ copies/ml, in source water $7.78 \times 10^3 \pm 7.13 \times 10^3$ copies/ml, and in finished water $6.91 \times 10^2 \pm 6.79 \times 10^2$ copies/ml (Lu et al. 2018). As the absolute abundance of ARGs in the environment is an indicator to a possible intimidation to community health, it is recommended that it should also be estimated and studied for PAH-contaminated environments.

Prospect of genetic transfer of ARGs in PAH-contaminated soil

It is known that once a resistance trait is selected, host bacteria can transfer genes between individual cells, creating an enriched resistome (i.e., the collection of latent and active resistance attributes in a community). Therefore, there is a high possibility that resistance can spread among the bacteria once a gene enters a system, either vertically or horizontally. ARGs in the PAH-contaminated soil was found to be located on the chromosome of PAH-resistant bacteria (Chen et al. 2017), discouraging the role of plasmids in the transfer. Busch et al. (2018) suggested the role of a two-component regulatory system (TCS) that serves as a stimulus-response coupling mechanism to allow organisms to sense and respond in PAH stress conditions (Busch et al. 2018). So, there is a possibility that ARGs are disseminated in the presence of stress factors, such as PAH contamination in soil, which ultimately select bacterial populations with enhanced survivability (Fig. 1).

Furthermore, it was also found that *MexE-MexF-OprN*, tripartite efflux pumps, might play a role in the quorum sensing, virulence, and detoxification and enriching ARGs (Das et al. 2020). In another report, it has been suggested that in *P. aeruginosa*, three-component efflux systems such as *MexE-MexF-OprN* and *MexC-MexD-oprJ* also contribute in quorum sensing (QS) response along with release of toxic metabolites (Strateva and Yordanov 2009). Furthermore, *CzcRS*, a two-component system of *P. aeruginosa*, increases the expression of the *czcCBA* efflux that contributes resistance for heavy metals such as Zn, Cd, and Co, and also decrease the expression of porin protein—*OprD*—which has a prominent role in providing resistance against imipenem (Liu et al. 2015). Therefore, presence of these types of efflux system in the soil metagenome indicates that bacteria may sense and respond to PAH stress conditions.

Efflux pumps are known to discharge the hydrocarbon pollutants from the bacterial cells, and these had been homologous to multidrug resistance (MDR) efflux pump which have a role in ARGs (Martinez et al. 2009). However, the genes encoding the efflux pump are located in the MGE regulated through class 1 integrons, which has already been discussed to have a role in increased richness of ARGs and ARBs in the contaminated soils (Fig. 2) (Sun et al. 2015). MDR genes found in most of the bacteria and have multiple roles in the quorum sensing, detoxification of metabolites, virulence, and association with AMR (Martínez 2008). It was documented that *mexF* and *oprJ* were the most common genes that play a role in the three-component efflux systems such as MexE-MexF-OprN and MexC-MexD-oprJ in the mutant strains of *P. aeruginosa* (Strateva and Yordanov 2009).

It may be assumed that soil contaminated with PAHs could be the strong selective environment for ARBs and their ARGs, which could be transferred to the other pathogenic or environmental bacterial species. In contaminated soils, PAH-degrading and tolerant bacteria were predominant over others, and further characterization of these strains exhibited their strong abilities to resist antibiotics (Maurya et al. 2020) (Fig. 2). It has been recorded that PAHs have the mutagenic properties which enriching ARGs directly by triggering stress/repair systems or by changing the DNA composition (Sun et al. 2015; Busch et al. 2018) (Figs. 1 and 2). However, the co-exposure effects are complicated, yet the stressed bacteria could indeed trigger genetic exchanges, which may ultimately lead to AMR. It is significant to mention here that integrons occur in every environment, and able to exchange among the bacterial species. In addition to this, it is also important for the adaptation and evolution of the bacterial genome (Table 1). Integrons have been detected in the microbiomes of forest soils, riverine sediments, hot springs, plant surfaces, deep-sea sediments, desert soils, Antarctic soils, aquatic biofilms, and marine sediments (Gillings et al. 2009; Elsaied et al. 2011).

Kang et al. (2015) reported the linkage of pyrene, phenanthrene, and benzo[g,h,i]perylene with ampicillin-resistant pUC19 plasmids. Through the in vitro transcription studies, authors confirmed that the less frequency of transformation of *AmpR* in pUC19 results in PAH-repressed *AmpR* regulation to

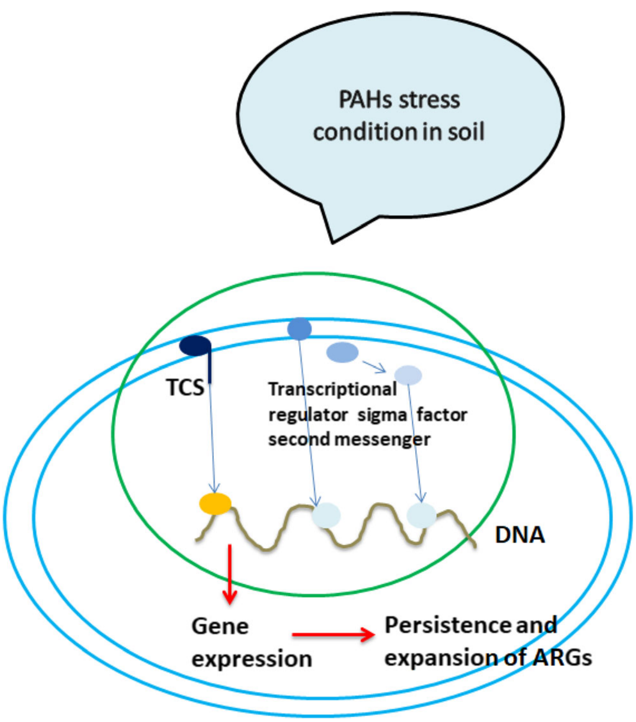

Fig. 1 Development and expression of ARGs under PAH stress condition: Two-component regulatory system (TCS) serves as a stimulus-response coupling mechanism to allow organisms to sense and respond under changes in PAH stress conditions: transmission of ARGs in the presence of PAH/antibiotic stress conditions, which finally select bacterial populations with higher survivability

RNA (Kang et al. 2015). Furthermore, Xie et al. (2016) found a progressive linkage among the presence of MGEs, and ARGs carrying β-lactam, tetracycline, vancomycin, and aminoglycoside resistance genes, and therefore, there is a high possibility of resistance by exchanging the ARGs between the microbial communities with the help of plasmids. However, half-lives of ARGs such as *sul1*, *tetA*, *tetW*, *ermB*, and *tetX* in the polluted soil were found considerably different, i.e., from weeks to months. In fact, in soil microbiome, ARGs and *int1* are present in the high concentration for long periods of time, and due to this, during the transportation of the organic wastes, chances of exposure of ARGs in the normal environment become high (Burch et al. 2014; Xie et al. 2016).

Rodgers et al. (2019) documented vertical transmission of ARGs through the selective pressures of PAHs/antibiotics in the soil microbiome. It was found that bacteria develop the ARGs in the polluted environment with enhanced survivability and replication (Rodgers et al. 2019). Several other reports have documented that exposure of naphthalene and phenanthrene associated with the conjugal transfer of ARGs present in class 1 integrons (Table 1) (Wang et al. 2017), allowing bacteria to adapt and evolve themselves in the diverse environments by acquiring the ARGs with improved gene regulation (Loot et al. 2017).

Reports are suggesting that some of the degradation pathways are also present on the same MGEs (e.g., catabolic plasmids of TOL, and NAH-7) that also carry the ARGs or heavy metal resistance genes. Therefore, it has been assumed that the extended exposure to antibiotics, metals, and/or PAHs might be associated with the extensive distribution of biodegradative abilities (Roy et al. 2002; Ben Said et al. 2008). Mathe et al. (2012) observed that ARGs and other resistance genes are usually expressed together, as PAHs or antibiotics or heavy metal contamination creates a selective environment for growth and expansion of bacteria. The authors reported that heavy metal resistance *P. fluorescens* BBN1, *P. corrugata* BBB2, *R. erythropolis* BGN2, *P. syringae* CSGN4, and *P. mandelii* CSGN1 found to be resistance to cefuroxime sodium, cefoperazone, cefotaxime, ceftazidime, cefaclor, moxalactam, tigecycline, piperacillin, amoxicillin, piperacillin-tazobactam, novobiocin, and nalidixic acid. Additionally, these isolates have the potential to degrade the aromatic and aliphatic hydrocarbons, i.e., naphthalene, benzene, toluene, xylene, and n-dodecane (Mathe et al. 2012). Ben Said et al. (2008) isolated *Pseudomonas putida*, *Shewanella oneidensis*, *Achromobacter xylosoxidans*, *Agrobacterium tumefaciens*, *Pseudomonas mendocina*, *Bacillus* sp., *Stenotrophomonas maltophilia*, *Pseudomonas nitroreducens*, *Shigella boydii*, *Staphylococcus* sp., *Acinetobacter* sp., and *Achromobacter xylosoxidans*, harboring PAHs degrading *idoA* and *nah*genes (Ben Said et al. 2008). Among these, 89% isolates found to be resistant to aminosid, beta-lactam, tetracycline, and co-trimoxazole antibiotics, while 8% of the isolates were resistant

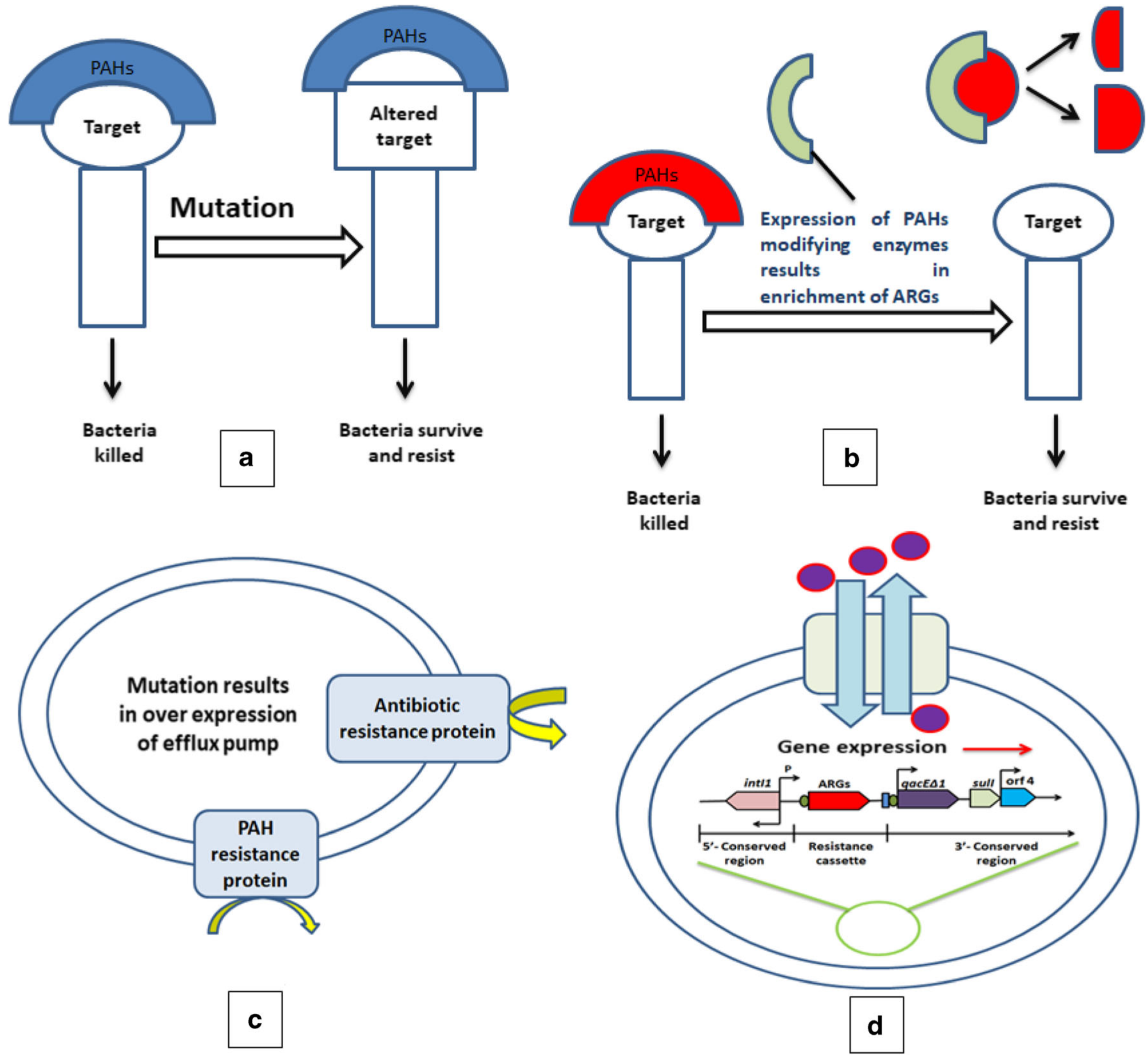

Fig. 2 **a** and **b** Mutations induced by PAHs by target alterations and/or by the help of inactivating enzymes. **c** PAHs and antibiotics act as a substrate for overexpression of the efflux pumps. **d** Overexpression of efflux pumps and exposure of PAH-mediated class I (*intI*) integron, allowing bacteria to develop and enrich the progression of ARGs with their gene expression

to penicillin, cefoxitin, amoxicillin, streptomycin, co-trimoxazole, neomycin, oxacillin, and tetracycline antibiotics. The isolates also showed resistance to heavy metals such as lead (Pb), copper (Cu), zinc (Zn), chromium (Cr), nickel (Ni), iron (Fe), and cobalt (Co) (Ben Said et al. 2008).

Studies on HGT between different microorganisms and factors that lead to the gene transfer in the environment are needed to be given due credit for the spread of ARGs in PAH-contaminated environments. Some authors have considered that antibiotic resistance mechanisms as a positive attribute related to resilience in strains proposed for bioremediation. Contaminated soils served as a selection pressure to soil microbial populations, and co-selection of resistance to pollutants and antibiotics has been determined. Bacterial populations with minimal antibiotic resistance along with pre-treatment of organic wastes to decrease selective pressure like antibiotic residues are necessary to evade environmental contamination with antibiotic-resistant bacteria and genes. The microbial community may probably be reduced in soils contaminated with hydrocarbons and such soils could be sensitive to the introduction of ARGs. Hence, the microbial population possessing multiple resistances traits including antibiotic resistance could be effective in hydrocarbon degradation (Cunningham et al. 2020a).

Recently, it had been reported that the metagenome of petroleum hydrocarbon–contaminated soil had two different MDR efflux pumps: *CmeABC* operon of *Campylobacter jejuni* and *MexE-MexF-OprN* (Das et al. 2020). These two efflux pumps comprising *RND*, *MexC*, *MexE*, *MdtB*, *MdtC*, *mdtD*, *MexD*, *MexE*, *MexF*, *MFS*, *MacB*, *BaeR*, *BaeS*, *CmeA*, *CmeB*, *CmeC*, *CmeR*, *MacA*, *MexT*, and *OprN* genes, were found to be involved in discharge of hydrocarbon pollutants from the bacterial cells. Interestingly, ARGs were highly enriched in such soil samples (Das et al. 2020). In earlier reports, RND efflux pumps are presumed to be responsible for the expression of multidrug resistance, and environmental adaptability (Martinez et al. 2009), and also involved in the PAH degradation (Hearn et al. 2003; Chain et al. 2006). *EmhABC* of *P. fluorescens* LP6a has a role in discharge of antibiotics and PAHs. It has also been reported that *EmhABC* efflux system act as a substrate for antibiotics and PAHs (Hearn et al. 2003; Adebusuyi and Foght 2011), while *P. putida* B6-2, RND efflux pump: *TtgABC*, showed higher expression levels and was contributing for resistance against ampicillin, cefotaxime, cefoxitin, nalidixic acid, novobiocin, tetracycline, chloramphenicol, erythromycin, gentamycin, and meropenem antibiotics (Yao et al. 2017).

Impact of metals and organic pollutants on abundance of ARGs

An anthropogenic activity not only pollutes the soil but also provides the opportunity for mixing between natural and indigenous microflora of soil with human-associated microbes. It has been reported that contaminated soils are the most favorable source of antibiotic, PAH, or heavy metal resistance genes (Gorovtsov et al. 2018). Therefore, bacteria linked with humans, animals, and soil indigenous flora interact in such environments, which is significant for co-occurrence of ARGs, as many of these bacteria linked with domestic animals or humans had been exposed to antibiotics, and henceforth also flourished under pressure associated with exposure of heavy metals or PAHs (Huijbers et al. 2019). In fact, mutation is not always involved in the co-selection process; however, HGT plays a significant role in the transfer of the resistance genes in organisms (Bengtsson-palme et al. 2018). There are two main motivating forces for co-selection of ARGs: co-resistance and cross-resistance (Maurya et al. 2020). Undeviating co-selection of ARGs in *Salmonella*, *Escherichia coli*, *Enterococcus faecalis*, and *Aeromonas salmonicida* subsp. *salmonicida* can be endorsed to multiple pollutants such as antibiotics, heavy metals, or PAHs (Seiler and Berendonk 2012; Maurya et al. 2020). Therefore, it may be hypothesized that the selection pressure created by the pollutants increases the secondary co-selection of ARGs, predominantly when many types of contaminants are present in the soil microbiome. Furthermore, it has also been noticed that linkage between the ARGs and heavy metal resistance genes is most common, as the resistance-carrying genes are usually present on the same MGEs (Chen et al. 2019; Maurya et al. 2020). However, higher concentrations of heavy metals in the soil microbiome could enforce the co-selection of ARGs by genetic couplings, where multiple types of contaminants is present (Baker-austin et al. 2006; Li et al. 2017; Yazdankhah et al. 2018). Hence, it could be speculated that for the impact of heavy metals, organic pollutants on abundance of ARGs quickly alters as per environmental surroundings or the selective pressure caused by the pollutants.

Challenges and prospects

The exact assessment of the influence of antibiotics on the diversity and activities of soil microbiome is an inordinate task for the researchers. Therefore, possible transfer of ARGs into the environmental bacteria may extremely prejudice the efficacy of antimicrobial therapy, and thereby posing a harmful health risk to the public health. Therefore, there is urgent need to identify and monitor the presence of ARGs in bacterial communities to elucidate the resistance mechanisms under the PAH stress conditions, to understand (1) how TCS serves as a stimulus-response coupling mechanism for enriching ARGs, (2) how mutations in efflux pumps affect enrichment of ARGs, and (3) how integrons are involved in dissemination and expansion of ARGs in diverse bacterial communities? Also, the extracellular processes for the interaction between ARG and organic pollutants may affect the ARG activity during HGT. Antibiotic resistance gene is a new type of pollutant, and its activity transformation mediated by common organic pollutants must be further studied and explained on the basis of current studies. Although aspects of combating antibiotic resistance remain essential, attention should be paid to the study of environmental enrichment and selection of the ARGs into the PAH-contaminated areas.

Conclusion

Research on studies of environmental enrichment as well as selection of ARGs is the ultimate means to overcome antibiotic resistance. Soils contaminated with PAHs are a good source of PAH-tolerant bacteria and their contamination had resulted in the shift in soil microbiomes which is also a rich source for resistance against antibiotics and endures multiple ARGs. Analysis of competition between the antibiotics and PAHs is essential for risk assessment. Exploring the influence of pollutant contamination would provide better understanding of the distribution of antibiotics and ARGs. Microbial communities may utilize a selective pressure for transmission of antibiotic resistance factors with occurrence of antibiotic residuals in the natural environment. Therefore, identification of microorganisms resistant to antibiotics should be emphasized, for monitoring the transfer of ARGs to other bacteria in the PAH-contaminated environments.

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Authors' contributions APM and JR: manuscript writing and compilation; PP: suggestions, comments, and final submission

Compliance with ethical standards

Consent for publication All authors gave their consent to publish the manuscript.

Competing interests The authors declare that they have no conflict of interest.

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